SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

*I **Mohammed Mudhassir A / 192424367** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Mudhassir
Signature of the student:

1, Box A: 6R, 4B, 2G → Total = 12
Box B: 5R, 7B, 3G → Total = 15
Box C: 8R, 2B, 5G → Total = 15
Selection probabilities:

- P(A)=1/4
- P(B)=1/2
- P(C)=1/4

Blue probabilities

$$P(B \mid A) = \frac{4}{12} = \frac{1}{3}$$
$$P(B \mid B) = \frac{7}{15}$$
$$P(B \mid C) = \frac{2}{15}$$

Using Bayes theorem,

$$P(C \mid Blue) = \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{7}{15} + \frac{1}{4} \times \frac{2}{15}}$$

Denominator

$$= \frac{1}{12} + \frac{7}{30} + \frac{1}{30} = \frac{7}{20}$$

Therefore

$$P(C \mid Blue) = \frac{1/30}{7/20} = \frac{2}{21}$$

Answer

$$\boxed{P(\text{Box C} \mid \text{Blue}) = \frac{2}{21} \approx 0.0952}$$

≈ 9.52%

2, Section A

$$P(B \mid A) = \frac{3}{10}$$

Section B

$$P(B \mid B) = \frac{6}{15} = \frac{2}{5}$$

Section C

$$P(B \mid C) = \frac{2}{15}$$

Selection probabilities

- A=1/4
- B=1/2

- C=1/4

Bayes theorem

$$P(C \mid Blue) = \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{2}{5} + \frac{1}{4} \times \frac{2}{15}}$$

Denominator

$$= \frac{3}{40} + \frac{1}{5} + \frac{1}{30} = \frac{37}{120}$$

Therefore

$$P(C \mid Blue) = \frac{1/30}{37/120} = \frac{4}{37}$$

Answer

$$\boxed{\frac{4}{37} \approx 0.1081}$$

≈10.81%

3, Total samples

$$1800 + 1800 = 3600$$

Testing (20%)

$$3600 \times 0.2 = 720$$

Each class

$$360$$

Given
TP=210
TN=190
FP=30
FN

$$= 360 - 210 = 150$$

Accuracy

$$= \frac{210 + 190}{720} = 55.56\%$$

Precision

$$= \frac{210}{210 + 30} = 87.5\%$$

Recall

$$= \frac{210}{210 + 150} = 58.33\%$$

Specificity

$$= \frac{190}{190 + 30} = 86.36\%$$

Answer
FN = **150**
Accuracy = **55.56%**
Precision = **87.50%**
Recall = **58.33%**
Specificity = **86.36%**
Model has **high precision and specificity but low recall**, meaning many malignant cases are missed.

4, Validation loss is minimum at **Epoch 3**.
Starts increasing from **Epoch 4**.
Answers
i)
Overfitting starts at Epoch 4
ii)

- Early Stopping
- Dropout/L2 Regularization

iii)
Overfitting causes excellent performance on training data but poor performance on unseen data because the model memorizes training examples instead of learning general patterns.

5, Validation accuracy peaks
Epoch 5 =73%
Then decreases.
Answers
i)
Overfitting starts after Epoch 5 (Epoch 6).
ii)
Training accuracy increases because the model memorizes training data, while validation accuracy decreases because it cannot generalize to unseen data.
iii)

- Early Stopping
- Data Augmentation (or Dropout)

6, Validation loss minimum
Epoch 4 =2.0
Answers
i)
Optimal generalization
Epoch 4
ii)
Overfitting signs

- Training loss decreases continuously.
- Validation loss increases after Epoch 4.

iii)
Regularization methods

- Dropout

- L2 Regularization (Weight Decay)

Both reduce overfitting by preventing the model from becoming overly complex.

7, TP=119
FN=6
TN=120
FP=5
Total=250
Accuracy

$$=\frac{239}{250}=95.6\%$$

Precision

$$=\frac{119}{124}=95.97\%$$

Recall

$$=\frac{119}{125}=95.2\%$$

Specificity

$$=\frac{120}{125}=96.0\%$$

F1

$$=95.58\%$$

Answer
Accuracy = **95.60%**
Precision = **95.97%**
Recall = **95.20%**
Specificity = **96.00%**
F1-score = **95.58%**

8, TP=138
FN=12
TN=135
FP=15
Accuracy

$$91\%$$

Precision

$$90.20\%$$

Recall

$$92.00\%$$

Specificity

$$90.00\%$$

F1

91.09%

Answer
Accuracy = **91.00%**
Precision = **90.20%**
Recall = **92.00%**
Specificity = **90.00%**
F1-score = **91.09%**

9, **TP=180**
FN=20
TN=170
FP=30
Accuracy
87.5%
Precision
85.71%
Recall
90%
Specificity
85%
F1
87.80%

10, TP=230
FN=20
TN=225
FP=25
Accuracy
95%
Precision
90.20%
Recall
92%
Specificity
90%
F1
91.09%

11, Shop X
Adulterated
30/50=3/5
Shop Y
20/60=1/3
Selection
X=1/3
Y=2/3
Bayes

$$P(Y \mid Adulterated) = \frac{\frac{2}{3} \times \frac{1}{3}}{\frac{1}{3} \times \frac{3}{5} + \frac{2}{3} \times \frac{1}{3}} = \frac{5}{8}$$

$$\boxed{\frac{5}{8} = 62.5\%}$$

12, Section A
15/50=3/10
Section B
25/50=1/2
Selection
A=3/4
B=1/4
Bayes

$$P(A \mid Defective) = \frac{\frac{3}{4} \times \frac{3}{10}}{\frac{3}{4} \times \frac{3}{10} + \frac{1}{4} \times \frac{1}{2}} = \frac{9}{14}$$

Answer

$$\boxed{\frac{9}{14} \approx 64.29\%}$$

13, Each class
1000
Testing=20%
Positive=200
Negative=200
TP=120
TN=150
FP
=50
FN
=80
Accuracy
67.5%
Precision
70.59%
Recall
60%
Specificity
75%

14, Each class
1200
Testing
25%
Positive=300
Negative=300

TP=180
TN=160
FP=140
FN=120
Accuracy
56.67%
Precision
56.25%
Recall
60%
Specificity
53.33%

15, Validation loss
2.6
2.2
1.9
1.8 ← minimum
1.9
2.1
2.4
2.7
Answers
a)
Best generalization: Epoch 4
b)
Overfitting begins at Epoch 5, because training loss continues decreasing while validation loss starts increasing.
c)
Two techniques:

- **Early Stopping:** Stops training when validation loss begins to increase.
- **Dropout (or L2 Regularization):** Prevents the model from memorizing training data, improving generalization.